

Dylan Allen

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PROFESSIONAL SUMMARY

Senior AI Engineer with 8+ years building production web applications and, for the past four, the LLM systems inside them. Full stack in Python, Java, TypeScript, React, and SQL, with deep recent focus on OpenAI-based systems, including prompt design, retrieval, output validation, and evaluation. Owns features end to end, from schema and API to shipped interface, and specializes in finding where a model gets things wrong and building the checks that catch it before users do. Proven in healthcare and enterprise environments where correctness, privacy, and reliability matter more than novelty.

SKILLS

AI & LLM Systems: OpenAI API, Prompt Engineering, Retrieval Augmented Generation (embeddings, vector search), LLM Evaluation & Rubric Design, Response Ranking, Hallucination Detection, Output Validation & Guardrails, Human-in-the-Loop Review, Responsible AI, AI Workflow Automation

Languages: Python, Java, TypeScript, JavaScript, SQL

Backend & APIs: FastAPI, Node.js, REST API Design, Microservices, ETL & Data Pipelines, Third-Party Integrations

Frontend: React, Next.js, Dashboards & Data Visualization

Data: PostgreSQL, MySQL, Schema Design, Query Optimization, Document Analysis & Data Extraction

Cloud & DevOps: AWS, Docker, CI/CD, Git / GitHub

Practices & Compliance: HIPAA, RBAC & Audit Logging, Agile / Scrum, Code Review, Unit Testing, Debugging

EXPERIENCE

Senior AI Engineer

Jun 2024 – Present

Akveo

Austin, TX

- Designed and executed evaluation frameworks for LLM responses, scoring accuracy, clarity, relevance, completeness, instruction following, and safety across enterprise business and technical use cases.
- Developed and iterated prompts to test model behavior against realistic business scenarios, including process optimization, reporting, workflow automation, data analysis, and software support tasks.
- Built scoring rubrics and structured evaluation criteria that improved consistency across repeated tasks and made model quality measurable rather than subjective.
- Performed side-by-side comparison and preference ranking of competing model responses, selecting stronger outputs on correctness, completeness, reasoning quality, and end-user value.
- Identified and documented hallucinations, unsupported claims, faulty reasoning chains, incorrect business logic, and formatting failures in model-generated output.
- Analyzed failures across large evaluation volumes to surface recurring patterns, then recommended changes to prompt design, response structure, and evaluation criteria.
- Wrote detailed technical feedback and error annotations to guide model improvement and fine-tuning, applying structured annotation guidelines to keep evaluations consistent and reproducible.
- Advanced responsible AI practice by testing outputs for safety, bias, and reliability before production use.

Software Engineer

Oct 2022 – May 2024

Orangesoft

Sheridan, WY

- Developed and deployed production LLM features using the OpenAI API and GPT models, including document analysis, structured data extraction, conversational AI assistants, NLP, text generation, and AI workflow automation.
- Applied prompt engineering as a repeatable process by building evaluation sets of known inputs and expected outputs, running A/B comparisons across prompt versions, and measuring accuracy before deploying changes to production.
- Implemented Retrieval Augmented Generation (RAG) pipelines with embeddings and vector search over client document sets, improving factual accuracy, reducing hallucinations, and making AI responses traceable to source documents.
- Engineered backend guardrails for LLM output, including JSON schema enforcement, field-level validation, confidence scoring, and human-in-the-loop review routing, so model-generated data met accuracy requirements before database persistence.
- Optimized AI system cost and latency through per-task model selection, prompt and context window reduction, response caching, and asynchronous background processing.
- Designed and developed RESTful APIs and backend services in Python, Java, and Node.js for user management, clinical workflows, reporting, data processing, ETL pipelines, scheduled jobs, and third-party API integrations.
- Built responsive front ends in React, TypeScript, and JavaScript for AI output review, chat interfaces, multistep forms, dashboards, and data visualization.
- Delivered software in a HIPAA-aware healthcare environment, implementing RBAC, audit logging, least-privilege data access, and secure handling of PHI.

Backend Developer

Jun 2018 – Oct 2022

Geomotiv

Alexandria, VA

- Designed and developed backend services and AI-powered features for web applications using Python, FastAPI, OpenAI APIs, and modern JavaScript frameworks.
- Built LLM workflows for chatbot support, document analysis, text generation, data extraction, and process automation.
- Implemented RAG and prompt engineering practices, testing against real user scenarios and edge cases, then refining prompts, logic, and validation rules from results.
- Engineered reusable backend services and API endpoints for AI features, covering request handling, response formatting, error handling, and logging.
- Integrated AI services with backend APIs, relational databases, and frontend applications to support end-to-end user experiences.
- Partnered with product managers, designers, and developers to translate business needs into practical AI solutions.

PROJECTS

LLM Evaluation & Scoring Framework

Jun 2024 – Present

Akveo

- Built rubric-based evaluation criteria and annotation guidelines for scoring LLM responses on accuracy, completeness, instruction following, and safety across enterprise business and technical use cases.
- Ran side-by-side preference ranking and failure pattern analysis across large evaluation volumes, feeding findings directly into prompt redesign and fine-tuning feedback.
- Built with: Python, OpenAI API, Structured rubrics, Annotation tooling, Responsible AI testing

RAG Document Intelligence Pipeline

2023 – 2024

Orangesoft

- Built a Retrieval Augmented Generation pipeline with embeddings and vector search over client document sets so every answer is traceable to its source document.
- Added structured data extraction with JSON schema enforcement, field-level validation, confidence scoring, and human-in-the-loop review routing before database persistence.
- Reduced cost and latency through per-task model selection, context window trimming, response caching, and asynchronous processing, all delivered in a HIPAA-aware environment.
- Built with: Python, FastAPI, OpenAI API, Embeddings & vector search, PostgreSQL, React / TypeScript

Conversational AI Assistant & Workflow Automation

2021 – 2022

Geomotiv

- Built LLM-based chatbot support, document analysis, and data extraction workflows on the OpenAI API, exposed through reusable FastAPI endpoints and integrated with backend services and databases.
- Established a prompt iteration and evaluation practice by testing against real user scenarios and edge cases, then refining prompts, logic, and validation rules from the results.
- Built with: Python, FastAPI, OpenAI API, JavaScript, REST APIs

EDUCATION

University of Mississippi

B.S., Computer Science.

2018